SYDE572 – Assignment 3

PCA

Part 1: PCA by hand
Consider the following dataset, containing 5 examples with 2 features each:

x_1	x_2
-2	-1
-1	0
0	1
1	2
2	3

Recall that with PCA we usually assume that the PCs are normalized (||w|| = 1), we need to center the data before we apply PCA, and that the direction of the first PC is the one that minimizes the orthogonal distance to all data points.

1. What is the first principal component?

2. What is the (L2-norm) reconstruction error of the point (3,3)? (Show your work.)

3. What is the (L2-norm) reconstruction error of the point (3,4)? (Show your work.)

Part 2: Data Visualization

Load the Iris dataset. https://archive.ics.uci.edu/dataset/53/iris
Marginalize (projective data onto the axis or remove the other feature dimensions) and visualize the dataset for any one feature, a combination of 2 features or a combination of 3 features. Show your results, it is arbitrary what axes you choose. Do any of these marginalization's highlight the classification of the dataset into nice clusters, if so, which ones. Comment.

Part 3: PCA

Load the Iris dataset. https://archive.ics.uci.edu/dataset/53/iris

Perform PCA on this dataset, you can use scikit-learn python functions. Visualize the results marginalized on the major axis, the top 2 axes, the top 3 axes. Provide the vectors that describe the PCA axes.

Part 4: Data Compression

It is important to know how much of the information in our dataset is captured by the low-dimensional PCA representation. In class we discussed the \analysis" view that PCA maximizes the variance that is explained by the PCs, and the connection between the Frobenius norm and the variance of a centered data matrix X.
Use this connection to answer the following:

1. How much of the variance is explained by our two-dimensional representation from the previous question?
2. How many PCs are required to explain 50% of the variance in the data?

Part 5: Non-linear Components

Euclidean distances between very different plants are unlikely to be particularly meaningful. However, since related plants tend to share similar traits we might expect the plants to live on a low-dimensional manifold.
This suggests that ISOMAP may give a better visualization. Use the python scikit-learn isomer function which computes the approximate geodesic distance (shortest path through a graph where the edges are only between nodes that are k-nearest neighbour) between each pair of points, and then fits a standard MDS (MultiDimensional Scaling) model using gradient descent. Hand in your code and the plot of the result when using the 3-nearest neighbours.
Compare the dimensionality reduction using the non linear function and PCA, which is preferable and why? Evaluate against data compression criteria or the ability to classify the data.

Grading Rubric

Each part is worth 5 marks. It will be graded on correctness, presentation style, and analytics (answering questions and the detail provided in the response).

This assignment will be graded out of **25** points, as follows:

For analytical results: /10. (A+ is 9.5)

- **A. (8-10)**
- **B (7-8)**
- **C (6-7)**

- **D (5)**
- **F (3)**

For presentation of data: \5

- **4.5 excellent**
- **4 good quality**
- **3 average**
- **2 below average**

Use guidelines for assignment 1 for submission